

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Adaptive Mobile E-Learning Platform Based on Quality of Experience (QoE)

1. Introduction

1.1 Purpose of the System

The purpose of this document is to specify the software requirements for the Adaptive Mobile E-Learning Platform based on Quality of Experience (QoE). The system is being developed to address the challenges of online education in low-resource environments, where unstable internet connectivity, high mobile data costs, and limited hardware restrict learners from accessing quality education.

This SRS serves as the authoritative reference for all functional behavior, performance expectations, constraints, and design guidance for the development team, testers, instructors, students, and all other stakeholders involved in the project.

1.2 Problem Statement

Online education platforms currently available are designed for high-bandwidth, stable internet environments. Students and instructors in regions like Cameroon face significant barriers:

- Unstable 3G/4G connectivity that causes frequent session interruptions
- High mobile data costs that discourage the use of video-heavy platforms
- Low-end smartphones with limited processing power, memory, and storage
- Frequent power outages that cut sessions short and drain device batteries
- No existing platform that automatically adapts content format to match real-time network conditions

As a result, students are forced to either skip learning sessions entirely or struggle through poor-quality, interrupted streams. There is a clear need for a platform that treats poor connectivity as the norm and adapts accordingly.

1.3 Objectives

The project aims to achieve the following objectives:

1. Design and implement an adaptive mobile e-learning platform that dynamically adjusts content delivery based on real-time network quality.
2. Provide seamless switching between video, audio-only, and text modes based on measured bandwidth, latency, and packet loss.
3. Enable offline access to downloaded learning materials so students can study without an active internet connection.
4. Monitor and record Quality of Experience (QoE) metrics throughout each session to support evaluation and improvement.

5. Build a lightweight, battery-efficient application compatible with low-end Android and iOS devices.
6. Support both English and French interfaces to serve Cameroon's bilingual education environment.

1.4 Scope

What the system will do: The system will deliver educational content — including video lectures, audio recordings, notes, quizzes, and downloadable files — to enrolled students and instructors. It will monitor network conditions in real time and automatically adapt the content delivery format to maintain the best possible learning experience given the available bandwidth.

What the system will not do: The system will not replace existing institutional management systems such as SIGE or Moodle. It will not handle financial transactions, course enrollment fees, or student grading at the institutional level. Advanced machine-learning-based personalization and live adaptive streaming are out of scope for Phase 1 and are planned for future versions.

2. Requirement Review and Analysis

This section reviews the requirements gathered in Task 2 through student surveys, instructor interviews, team brainstorming sessions, and reverse engineering of existing e-learning platforms. The review evaluates whether the collected requirements are complete, clear, technically feasible, and logically connected.

2.1 Completeness Analysis

The gathered requirements were assessed to confirm they adequately cover all key stakeholder groups and system domains. The following areas were found to be sufficiently addressed:

- User authentication and account management
- Access to educational materials including videos, notes, and quizzes
- Support for both live and recorded learning sessions
- Adaptive content delivery under unstable network conditions
- Dynamic video quality adjustment
- Automatic switching between video and audio-only modes
- QoE monitoring and evaluation
- Low-data content delivery strategies
- Accessibility and usability considerations

The requirement gathering process also captured the major environmental challenges in the target environment:

- Unstable internet connectivity and frequent session interruptions
- High mobile data costs relative to income levels in the region
- Use of low-end smartphones with limited memory and storage

The following gaps were identified and have been addressed in this document:

Gap Identified	How It Was Addressed
Accessibility standards not specified	WCAG 2.1 Level AA compliance added to non-functional requirements.

Gap Identified	How It Was Addressed
Offline synchronization scope unclear	Offline download, quiz submission, and background sync requirements added to Section 3.
Security standards not explicit	TLS encryption, bcrypt password hashing, and data anonymization added to Section 4.
Bilingual interface not mentioned	English and French interface support added to usability requirements.

2.2 Clarity Analysis

Several requirements collected in Task 2 were too vague to be directly implemented or tested. These were refined into specific, measurable statements as shown below:

Original (Vague) Statement	Refined Requirement
Improve online learning experience	The system shall dynamically adjust video quality based on real-time bandwidth, choosing from at least three resolution levels (e.g., 1080p, 480p, 144p).
Reduce interruptions during classes	The system shall automatically switch from video to audio-only mode when measured bandwidth drops below a set threshold, without user action.
Support weak internet users	The system shall provide downloadable learning materials for offline access, supporting pause and resume across connection interruptions.
Handle poor network conditions	The system shall continuously monitor bandwidth, latency, and packet loss, and apply adaptation rules within three seconds of detecting a change.
The app should be easy to use	Any main feature shall be reachable within three taps from the home screen. A first-time user shall start their first lesson within five minutes.

Key technical terms — including Quality of Experience (QoE), adaptive streaming, bandwidth, latency, and packet loss — have been clearly defined in Section 1.5 and used consistently throughout the document.

2.3 Technical Feasibility Analysis

Each major system feature was evaluated against current technologies to determine whether it can realistically be implemented within the project scope and timeline.

Feature	Feasibility	Technical Notes
User authentication	High	JWT-based login with encrypted credential storage via OS keychain. Well-understood and widely implemented.
Course content delivery	High	REST APIs for content access; ExoPlayer (Android) / AVPlayer (iOS) for video playback; CDN for performance.
Adaptive streaming (pre-recorded)	High	HLS or MPEG-DASH with multiple server-encoded resolutions. This is Phase 1 scope. Well-supported by existing tools.

Feature	Feasibility	Technical Notes
Adaptive streaming (live)	Moderate	Requires WebRTC simulcast and an SFU media server (e.g., mediasoup). Technically feasible but adds cost and complexity. Planned for Phase 2.
Audio-only fallback	High	Simple bandwidth threshold trigger. Low complexity. No significant additional infrastructure required.
QoE monitoring	High	Android NetworkCapabilities API; iOS NWPathMonitor; WebRTC statistics for live sessions.
Content recommendation	Moderate	Rule-based recommendation is feasible from the start. Collaborative filtering requires sufficient historical data. Planned for Phase 2.
Offline access and sync	Moderate	Room (Android) / Core Data (iOS) for local storage. Background sync with conflict resolution adds moderate complexity.
Mobile compatibility (low-end)	High	Achievable if the application remains lightweight and computation-heavy tasks run server-side.

2.4 Dependency Relationship Analysis

Some features cannot function without other system components being in place. The table below maps the key dependencies to guide the development sequence:

Requirement	Depends On	Dependency Type
Adaptive video streaming	Real-time network monitoring	Mandatory — adaptation needs measured input.
Audio-only fallback	Bandwidth threshold detection	Mandatory — fallback is triggered by a measured value.
QoE evaluation	User activity logging + network monitoring	Mandatory — QoE is a combination of both.
Recommendation system	User interaction analytics	Mandatory — recommendations need historical data.
Live lecture adaptation	Stable server communication	Mandatory — adaptation requires server-side encoding control.
Downloadable resources	Content management system	Mandatory — download requires existing content storage.
Student dashboard	User activity logging	Mandatory — dashboard data comes from session logs.
Offline sync	Content download module	Mandatory — sync requires locally stored content.

Network monitoring is the most critical foundational component. Multiple adaptive features depend directly on it. This layered dependency structure will guide the system architecture in Task 4.

3. Description of the Existing System

Currently, online learning in the target environment is carried out through a combination of informal and formal methods that were not designed for low-resource conditions.

3.1 How the Current System Works

Students and instructors primarily rely on platforms such as WhatsApp, Google Classroom, and YouTube for content sharing. Instructors upload documents and videos to these platforms; students download or stream them manually. Live sessions are conducted via Zoom or Google Meet. There is no central platform that integrates content management, communication, and quality adaptation.

3.2 Limitations of the Existing System

- No automatic quality adaptation — platforms stream at a fixed quality regardless of the user's network conditions.
- High data consumption — video-heavy platforms like YouTube and Zoom consume far more data than students can afford.
- No centralized management — course materials are scattered across multiple platforms with no unified access point.
- No QoE monitoring — instructors have no visibility into how students are experiencing content delivery.
- No offline-first design — most platforms require an active internet connection to function at all.
- Poor performance on low-end devices — video conferencing tools in particular demand more processing power than budget smartphones can provide.

3.3 Why a New System Is Needed

The limitations above directly harm student learning outcomes. A platform that monitors network quality in real time and adapts content delivery automatically — without requiring any action from the student — is necessary to make online education genuinely accessible in low-resource settings. The proposed system replaces the fragmented, non-adaptive collection of existing tools with a single, purpose-built, QoE-aware learning environment.

4. Proposed System

4.1 Overview

The proposed system is an adaptive mobile e-learning platform that delivers educational content to students and instructors while continuously monitoring network quality. When network conditions change, the system automatically switches the content delivery format — from high-quality video to lower-resolution video, audio-only, or text — without interrupting the session or requiring any action from the user.

4.2 Key Features

- Adaptive bitrate video streaming with automatic quality switching based on real-time bandwidth
- Automatic fallback to audio-only mode when video is not sustainable
- Offline content download with background sync when connectivity returns
- QoE monitoring dashboard for instructors showing per-student session quality

- Role-based access control for students, instructors, and administrators
- Live lecture support with chat and automatic session recording
- Student progress tracking and course analytics
- Lightweight mobile application compatible with low-end Android and iOS devices

4.3 Advantages Over the Existing System

Existing System Limitation	How the Proposed System Addresses It
No quality adaptation	Automatic adaptive bitrate streaming adjusts quality every few seconds based on measured bandwidth.
High data consumption	Audio-only fallback and local caching reduce data usage significantly. Usage targets: ≤ 50 MB/hr audio, ≤ 200 MB/hr video.
No centralized management	A single platform integrates content upload, delivery, live sessions, and analytics.
No QoE visibility for instructors	Real-time and historical QoE dashboards give instructors session quality data per student.
No offline access	Students can download content for offline study; progress syncs automatically when reconnected.
Poor low-end device performance	Heavy processing runs server-side; the client application is kept lightweight for budget smartphones.

4.4 Technologies to Be Used

Component	Technology
Mobile (Android)	Native Android (Java/Kotlin) or React Native
Mobile (iOS)	Native iOS (Swift) or React Native
Video streaming	HLS / MPEG-DASH adaptive bitrate streaming
Live sessions	WebRTC (Phase 2: SFU via mediasoup or Janus)
Backend API	Node.js / Django REST Framework
Database	PostgreSQL (relational data) + Firebase (real-time events)
Offline storage	Room (Android) / Core Data (iOS)
Authentication	JWT / OAuth 2.0
Network monitoring	Android NetworkCapabilities API / iOS NWPathMonitor
Cloud infrastructure	AWS Educate / Google Cloud for Education (Phase 1)

5. Functional Requirements

Functional requirements describe what the system must do. Each requirement uses "The system shall..." to ensure clarity and testability. Requirements are numbered for traceability.

5.1 User Authentication and Account Management

- FR-01: The system shall allow students and instructors to register with a valid email address and a secure password.
- FR-02: The system shall verify registered users with encrypted credential checks before granting access.
- FR-03: The system shall assign different access permissions to students, instructors, and administrators.
- FR-04: The system shall support password recovery through a verified email-based reset process.
- FR-05: The system shall automatically terminate sessions that have been inactive for longer than a configurable timeout period.

5.2 Course Management

- FR-06: The system shall allow instructors to upload course materials including videos, notes, quizzes, and downloadable files.
- FR-07: The system shall allow instructors to organize courses by subject or module.
- FR-08: The system shall allow instructors to edit and delete their uploaded materials.
- FR-09: The system shall provide a searchable course catalogue allowing students to find content by title, subject, or instructor.
- FR-10: The system shall allow instructors to set a preferred delivery mode (video, audio, or text) per lesson.

5.3 Content Delivery

- FR-11: The system shall provide access to both live and pre-recorded lectures.
- FR-12: The system shall stream educational videos to users using adaptive bitrate protocols.
- FR-13: The system shall allow students to download learning materials for offline access.
- FR-14: The system shall support pause and resume of content downloads across connection interruptions.

5.4 Adaptive Learning Features

- FR-15: The system shall measure available network bandwidth at least once every five seconds during content playback.
- FR-16: The system shall automatically switch video quality — choosing from at least three resolution levels — based on measured bandwidth.
- FR-17: The system shall automatically switch from video to audio-only mode when bandwidth falls below a defined minimum threshold.
- FR-18: The system shall revert from audio-only to video automatically when bandwidth recovers above the threshold, without user action.
- FR-19: The system shall display on-screen subtitles as a fallback option when audio quality is poor.
- FR-20: The system shall apply all content delivery adaptation decisions within three seconds of detecting a sustained network change.

5.5 QoE Monitoring

- FR-21: The system shall continuously measure and record bandwidth (kbps), latency (ms), packet loss (%), and jitter (ms) during sessions.

- FR-22: The system shall log all content adaptation events with timestamps and the network conditions that triggered them.
- FR-23: The system shall record user interaction events including play, pause, quiz completion, and navigation.
- FR-24: The system shall provide instructors with a QoE dashboard showing average session quality per student and per course.

5.6 Live Lecture Support

- FR-25: The system shall support live video lecture sessions that instructors can start and students can join.
- FR-26: The system shall include a real-time text chat feature during live sessions.
- FR-27: The system shall automatically record live sessions and make recordings available to enrolled students after the session ends.

5.7 Student Progress and Analytics

- FR-28: The system shall maintain a persistent student dashboard showing completed modules, quiz scores, and time spent per course.
- FR-29: The system shall send in-app notifications about upcoming deadlines, new content, and instructor announcements.
- FR-30: The system shall generate engagement and completion analytics reports for instructors per course.

5.8 Administration

- FR-31: The system shall allow administrators to create, suspend, and delete user accounts for all roles.
- FR-32: The system shall provide administrators with a system-wide monitoring view showing server load, active sessions, and error logs.
- FR-33: The system shall validate all data entered during registration and content submission to prevent incomplete or duplicate entries.

6. Non-Functional Requirements

Non-functional requirements define the quality standards the system must meet. These are as important as functional requirements and must be verified through testing or measurement.

6.1 Performance

- NFR-01: The system shall apply content delivery quality changes within three seconds of detecting a sustained change in network conditions.
- NFR-02: The system shall support at least 200 concurrent users without a noticeable degradation in response time.
- NFR-03: Course content pages shall load within four seconds on a 3G connection (approximately 1 Mbps).
- NFR-04: Students shall be able to join a live session within eight seconds of pressing the join button.

6.2 Reliability

- NFR-05: The system shall maintain an uptime of at least 99.5% per month, excluding planned maintenance windows.
- NFR-06: If adaptive streaming fails, the system shall fall back to a low-quality static stream rather than ending the session.
- NFR-07: Content downloads shall support pause and resume without restarting after a connection interruption.
- NFR-08: User actions occurring during a brief connection loss shall be saved locally and synced automatically when connectivity returns.

6.3 Security

- NFR-09: All data transmitted between the application and server shall be encrypted using TLS 1.2 or higher.
- NFR-10: User passwords shall be stored using bcrypt hashing with a minimum cost factor of 10.
- NFR-11: User interaction data used for analytics shall be anonymized before being stored.
- NFR-12: The system shall limit login attempts to prevent brute-force attacks.

6.4 Scalability

- NFR-13: The system architecture shall support scaling to ten times the initial user base without a full redesign.
- NFR-14: System modules shall be independently deployable so that the adaptation engine, content delivery, or analytics module can be updated without affecting others.
- NFR-15: All API endpoints shall use version numbers to allow updates without breaking existing app versions.

6.5 Efficiency

- NFR-16: The application shall not consume more than 50 MB of mobile data per hour in audio-only mode.
- NFR-17: The application shall not consume more than 200 MB of mobile data per hour when streaming standard-quality video (480p).

- NFR-18: Background QoE monitoring shall use event-based measurement to avoid unnecessary battery drain.
- NFR-19: The application shall cache previously accessed content locally to reduce redundant data downloads.

6.6 Usability and Accessibility

- NFR-20: A first-time user shall be able to find and start their first content module within five minutes of account creation, without external help.
- NFR-21: Any main feature shall be reachable within three taps from the home screen.
- NFR-22: Text content shall meet WCAG 2.1 Level AA minimum contrast requirements.
- NFR-23: The platform shall support both English and French language interfaces.
- NFR-24: The application shall function correctly on devices running Android 8.0 or later and iOS 13 or later.

7. User Stories

User stories describe the expected interactions between stakeholders and the system. Format: As a [user type], I want [feature], so that [benefit]. Stories are derived directly from the data gathered and trace to functional requirements in Section 5.

7.1 Student Stories

ID	User Story	Priority
US-01	As a student, I want video quality to adjust automatically so that I can keep learning even when my connection is weak.	High
US-02	As a student, I want the platform to switch to audio-only during poor connectivity so that lectures are not interrupted.	High
US-03	As a student, I want to download notes and materials so that I can study offline without using extra data.	High
US-04	As a student, I want access to low-resolution videos so that I can reduce my mobile data consumption.	Medium

7.2 Instructor Stories

ID	User Story	Priority
US-05	As an instructor, I want to upload materials in different formats so that students with different network conditions can access them.	High
US-06	As an instructor, I want to conduct live and recorded lectures so that learning remains flexible.	High
US-07	As an instructor, I want to see student participation data so that I can evaluate engagement.	Medium

7.3 Administrator Stories

ID	User Story	Priority
US-08	As an administrator, I want to monitor system performance and QoE metrics so that the platform stays reliable.	High
US-09	As an administrator, I want to manage user accounts and permissions so that the system remains secure.	High

8. System Constraints and Assumptions

8.1 Constraints

These are fixed limitations the platform must operate within. Every design decision must respect these constraints.

Constraint	Description	System Impact
Limited Connectivity	Many users face unstable 3G/4G connections with frequent drops and high latency.	Adaptive delivery, audio fallback, offline access, and session resumability are all required core features.
High Data Costs	Mobile data is expensive relative to average income levels in Cameroon.	The system must minimize data usage at every quality tier and support local content caching.
Low-End Devices	Many users have smartphones with limited RAM (1–2 GB), slow processors, and small storage.	The app must stay lightweight. Heavy computation must run server-side, not on-device.
Power Instability	Frequent power outages limit charging and battery life.	Background processes must use event-based monitoring. Sessions must support pause and resume.
Privacy and Security	The system collects personal user data and learning analytics.	All data must be encrypted in transit. Analytics data must be anonymized before storage.
Digital Literacy	User technical experience varies significantly across the student and instructor population.	The interface must be simple enough to navigate without training. Three-tap access to all main features.

8.2 Assumptions

- Users have access to at least a 2G/EDGE mobile data connection at minimum during sessions.
- Instructors have access to a moderately stable connection when uploading content.
- All users own or have access to a smartphone running Android 8.0 or later, or iOS 13 or later.
- The institution hosting the platform will provide or fund basic cloud server infrastructure.
- Users understand basic smartphone operation, including installing an application and navigating a menu.
- Course content rights belong to the instructors uploading them, and the institution has appropriate permissions.

9. User Interface Requirements

The user interface must be simple, lightweight, and usable on small screens common in the target environment. The following requirements define how the interface should look and behave.

9.1 General Interface Requirements

- The interface shall use a clean layout with clear visual hierarchy and minimal graphic elements that would increase data usage.
- Navigation shall use a bottom tab bar or side drawer, with main sections — Home, Courses, Live, Downloads, Profile — always visible.
- All interactive elements (buttons, links, input fields) shall be large enough to be comfortably tapped on a small screen (minimum 44x44 dp touch target).
- The interface shall support both portrait and landscape orientation.
- Font sizes shall meet WCAG 2.1 Level AA contrast requirements for readability.

9.2 Key Screens

Screen	Key Elements
Login / Register	Email field, password field, login button, register link, forgot password link.
Home / Dashboard	Enrolled courses list, recent activity, upcoming deadlines, quick access to downloads.
Course Page	Module list, progress indicator, content type toggle (video/audio/text), download button.
Video Player	Adaptive stream player, quality indicator, audio-only toggle button, subtitle toggle, progress bar.
Live Session	Video feed, chat panel, participant count, network quality indicator, leave button.
Student Progress	Completed modules, quiz scores, time spent, course completion percentage.
Instructor Dashboard	Course list, student QoE summary table, upload button, session start button.
Admin Panel	User management table, system load indicators, error log viewer, QoE report access.

9.3 Network Quality Indicator

The application shall display a persistent, unobtrusive network quality indicator on all content screens. This indicator shall show the current delivery mode (HD Video / SD Video / Audio Only / Text) so students are always aware of the active quality tier and can understand why the format may have changed.

10. Hardware and Software Requirements

10.1 Client (User Device) Requirements

Requirement	Minimum Specification
Operating System	Android 8.0 (Oreo) or later; iOS 13 or later
RAM	1 GB minimum (2 GB recommended for smoother video playback)
Storage	500 MB free for application installation; additional space for downloaded content
Processor	ARM Cortex-A53 (quad-core, 1.4 GHz) or equivalent
Network	2G/EDGE minimum; 3G or better recommended for video streaming
Screen	4-inch minimum display

10.2 Server Requirements

Requirement	Specification
Operating System	Ubuntu 22.04 LTS (or equivalent Linux distribution)
RAM	8 GB minimum for Phase 1; 16+ GB for live streaming
Storage	500 GB minimum for video content; scalable cloud storage (AWS S3 / Google Cloud Storage)
Processor	4-core CPU minimum; 8-core recommended for transcoding
Bandwidth	100 Mbps uplink minimum for Phase 1 deployment
Database	PostgreSQL 14 or later

10.3 Development Tools and Software

Category	Tools / Technologies
Mobile Development	Android Studio (Java/Kotlin); Xcode (Swift); or React Native (cross-platform)
Backend	Node.js or Django REST Framework; Express.js for API routing
Database	PostgreSQL; Firebase Realtime Database for live events
Video Streaming	FFmpeg for video transcoding; HLS.js / ExoPlayer for adaptive playback
Version Control	Git (GitHub or GitLab)
Testing	JUnit (Android); XCTest (iOS); Postman (API testing)
CI/CD	GitHub Actions or GitLab CI

11. Requirement Prioritization

Requirements are sorted into three priority levels to guide development. High priority items form the Minimum Viable Product (MVP) for Phase 1.

Priority	Requirements	Phase	Rationale
High	User authentication (FR-01 to FR-05), course content delivery (FR-11 to FR-14), adaptive streaming pre-recorded (FR-15 to FR-20), live adaptive streaming (SFU-based), QoE monitoring (FR-21 to FR-24)	Phase 1	Core features without which the platform cannot function.
Medium	Student dashboard (FR-28 to FR-30), live lecture basic (FR-25 to FR-27), instructor content management (FR-06 to FR-10), administration (FR-31 to FR-33)	Phase 2	Important for completeness and usability. Depends on Phase 1 stability.
Low	Advanced content recommendation, predictive analytics, extended personalization	Phase 3	Enhancements that require more data, infrastructure, and development time.

12. Testing Requirements

The following test cases define key validation criteria for the core system features.

Test ID	Feature Being Tested	Test Input / Condition	Expected Result
TC-01	User registration	Valid email and password submitted	Account created successfully; confirmation message shown.
TC-02	Login with wrong password	Correct email, incorrect password	Login denied; error message shown; attempt logged.
TC-03	Adaptive video quality switch	Bandwidth drops below defined threshold during playback	Video quality reduces automatically within 3 seconds, no session end.
TC-04	Audio-only fallback	Bandwidth drops below audio-only trigger threshold	Platform switches to audio-only mode without user action within 3 seconds.
TC-05	Offline content access	Student opens downloaded content with no internet	Content plays correctly from local storage; no network error shown.
TC-06	Offline sync	Student completes quiz offline, then reconnects	Quiz result syncs to server automatically in the background.

Test ID	Feature Being Tested	Test Input / Condition	Expected Result
TC-07	QoE logging	Session running with varying network conditions	Bandwidth, latency, packet loss logged at ≥ 1 entry per 5 seconds.
TC-08	Load test — 200 users	200 concurrent users streaming content simultaneously	All users receive content without service degradation.
TC-09	Session recovery	Network drops for 15 seconds during playback	Session resumes automatically when connection returns; no data lost.
TC-10	Three-tap navigation	New user opens the application	Any main feature is reachable within three taps from the home screen.

3. Conclusion

This Software Requirements Specification has transformed the raw data gathered in Task 2 into a comprehensive, structured, and validated set of requirements for the Adaptive Mobile E-Learning Platform based on Quality of Experience (QoE).

The completeness analysis confirmed that all key stakeholder areas are covered, with identified gaps addressed in this document. The clarity analysis refined vague statements into specific, testable requirements. The feasibility review confirmed that core features are achievable with current technology, with the key distinction that pre-recorded adaptive streaming is ready for Phase 1 while live adaptive streaming and advanced recommendation systems are planned for later phases.

The dependency analysis established the order in which system components must be built. The system constraints section clearly defined the real-world limitations — unstable connectivity, high data costs, low-end hardware, and battery constraints — that every design decision must respect. The prioritization table defines the MVP scope for Phase 1.

The resulting functional requirements (FR-01 to FR-33), non-functional requirements (NFR-01 to NFR-24), user interface specifications, hardware requirements, and testing criteria together provide a solid, complete foundation for the system design and implementation phases that follow.